Comp : ARHAM (+923351328979) | Dir: SAMUNA CH. | Co. : KAINAT CH.

OBJECTIVES

Lectures 98 to 112

1. Reason of death identify on the basis of **Post Mortem Report & Diagnostic Test**
2. In zoo full time officer **1**
3. Maintenance of hygienic condition by **Graveyard**
4. A master plan should be made for the next **5-25 Years**

Lectures 113 to 127

5. The name of argali is a … for sheep. **Mongolian Word**
6. Woolly flying squirrel native to **Pakistan**
7. Coat of panther is **Tawny or Light Yellow**
8. Snow leopard **Thick Fur**
9. Wild ass poaching for meal **Medicinal Purpose**

Lectures 128 to 143

10. Bhulan is the common name of **Indus Dolphin**
11. Indus dolphin navigation **Echolocation System**

Lectures 144 to 159

12.IUCN status of red lynx **Least Concern**
13.IUCN status of Indian vulture **Critically Endangered**
14.IUCN status of Indian spotted eagle **Vulnerable**
15. Bones of leopard cat are used in some traditional … medicine. **Asian**
16. Body color of jungle cat is **Sandy Grey**
17. Hog deer protected in **Bangladesh & India**
18. The deliberate movement of organisms from one site for … in another. **Release**

Lectures 144 to 159

19.IUCN status of brown bear **Least Concern**
20. Musk deer refer any one or all **7**



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21. In brown bear sexual maturity is from … of age. **4.5-7 Year**
22. Barking deer is little dainty creature & weights **22-25kg**
23. Urial subspecies locally called **Punjabi Dunba**

Lectures 160 to 187

24. Capra body length is …52 inches **4.4 Feet**
25. Foxes are **Omnivorous Mammals**
26. … throated marten larger in size **Yellow**
27. Civets are related to **Cats**

Lectures 188 to 201

28. Status of kingfisher **Least Concern**
29. IUCN status of houbara bustard **Vulnerable**
30. Sexes in koklass is **Dimorphic**
31. Kingfisher long term decline chemical & **Biological**
32. Egg of chukor hatched in 23 to … days. **25**

FROM MID

33. Wildlife is natural … with a great importance. **Renewable Resource**
34. Ayubia national park support … species of mammals. **31**
35. IUCN recognize … game reserve in Pakistan. **66**
36. Chashma & Taunsa Barrage dolphin sanctuary was declared open to the public **In 1972**
37. Hazarganji chiltan park established in **1978**
38. Elephant eat up to … of food per day. **450kg**
39. Rosy periwinkle leaves extracted as **Anticancer**
40. The largest park in Sindh **Kirthar National Park**
41. Endemic species **Galapagos Fauna**
42. Pinecones **Coniferous**
43. Criteria declared for wet land is **9**

EXTRA MCQs



Comp : ARHAM (+923351328979) | Dir: SAMUNA CH. | Co. : KAINAT CH.

44. Madras wild elephant preservation act **1873**

45. The SSC is governed by a steering committee which is headed by the … chair. **Commission**

46. **Schedule III** wild birds & animals protected throughout the year

47. … determination any wild life species found in any area. **Habitat**

48. Indus dolphin reserve is a natural wet land situated in the … **Sukkar**

49. Gir Forest National Park is a wildlife sanctuary in Gujarat, western India. It was established to protect --- **Asiatc Lions**

50. Largest National Park in Pakistan **Hingol National Park**

51. Second Largest National Park in Pakistan **Kirthar National Park**

52. Conifer trees (pine, spruce, fur, etc.) use their … **Pinecones**

53. The yellow throated marten is considerably … in size than the short marten. **Larger**

54. Panther coat ranges from tawny or light yellow in … dry habitats to … in dense forest. **Warm , Reddishorange**

55. Third largest national park in Pakistan **Khunjerab National Park**

56. In Pakistan presently there are … wildlife sanctuaries. IUCN recognized number of wildlife sanctuaries in Pakistan …. **99, 66**

57. Kilik/Mintaka Game Reserve lies along the border with … **China**

58. Astore Wildlife Sanctuary is located in Gilgit Baltistan … **Pakistan**

59. Data Deficient is therefore not a category of … **Threat, Lower Risk**

60. … is the human-mediated movement of living organisms from one area, with release another. **Translocation**

61. … have sharp, curved talons, used primarily for grasping prey. **Sakers**

62. The … looks like a miniature, long-legged and more-slender version of a leopard. **Leopard Cat**

63. … has been an important component of traditional medicine in Eastern Asia for millennia. **Bear Bile**

64. Yield of musk is only about … grams. **25**

65. The intentional movement & release of an organism inside its indigenous range from which it has disappeared. **Reintroduction**

66. The intentional movement & release of an organism into an existing population.



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Reinforcement

67. In the tropics, leopard cat fur is … with white underparts. **Yellowish Brown**
68. Jungle cat has tail about … of head & body length. **1/3**
69. Jungle cats feed mainly on prey that weighs less than … **1 Kg**
70. The bobcat is crepuscular, and is active mostly during … **Twilight**
71. The biggest threat to bobcat is its … **Fur**
72. The fishing cat is killed for consumption as a part of a … **Cultural Practice**
73. The fishing cat is largely nocturnal. Each individual eats b/w … & … rodents per year. **365,730**
74. The Asiatic Black Bear has been included on CITES Appendix I since … **1979**
75. The … is valued for its cosmetic & pharmaceutical properties. **Musk Deer**
76. Each antler of stag consists of … tines. **5**
77. Kashmir stag lives in groups of … individuals in dense riverine forests & high valleys. **2 To 18**
78. True wild sheep are included in the genus … **Ovis**
79. Gestation period of wild sheep **150-180 Days**
80. An antelope is a member of a number of … ungulated species. **Even-Toed**
81. Nilgai means **Blue Cow**
82. Breeding populations of nilgai exist only in … plantations. **Changa Manga & Lal Suhanra**
83. The young nilgai spend most of the … lying down & standing up only to suckle. **10 Days**
84. The black buck gets their name from the … attained on maturity at three years of age. **Black Pelage of Males**
85. Civets secrete from their anal glands, a greasy substance called “civet” which is used in making … **Perfumes**
86. Otters are … carnivores. **Aquatic**
87. Indian otter is found throughout the lower Indus riverine system of … **Punjab & Sindh**
88. Male peafowls are known as … which are members of the pheasant family.

Peacock

89. The peacock is the national bird of …. **India**
90. Pheasant is a large long tailed … bird. **Game**
91. The bird "koklass" is having a long central & two lateral crests on the head also called … **Pucras**
92. Tragopans are commonly called … **Horned Pheasants**
93. The word "tragos" means … **Goat**
94. Houbara bustard is a large sized … **Running Bird**
95. Breeding (sand dune cat) in the wild is seasonal with births born. **January – April**
96. Sexual maturity in sand cats is reached at about … months.**9-14**
97. In Pakistan, sand cats are reported in … only. **Baluchistan**
98. IUCN status of Panther is …. **Critically Endangered**
99. Lifespan of panther is …. **10 To 12 Years**
100. Leopards are sexually dimorphic as males tend to be … than females. **Larger**
101. Typically, female panthers stop reproducing around … years old. **8.5**
102. The primary threats to leopards are …. **Anthropogenic**
103. The Indian wild asses are significantly … than donkeys. **Larger**
104. The belly, buttocks & muzzle of asses are …. **White**
105. The Asiatic wild ass is included in CITES: **Appendix I**
106. … is one of the nine Argali subspecies. **Marco Polo Sheep**
107. … of wild sheep have two large corkscrew shaped horns. **Males**
108. Females of Marco polo sheep are sexually mature at: **2 Years**
109. Males of Marco polo sheep may not sexually mature until:**5 Years**
110. Argali are included on Appendix … of CITES. **I & Ii**
111. Great Indian Bustard is included in … of CITES. **Appendix I**
112. In … dolphins were protected under the Wildlife Act of Sindh. **1972**
113. In 1974 the government of Sindh declared the … b/w the Sukkur & Guddu Barrages a dolphin reserve. **Indus River**
114. The government of Punjab prohibited deliberate killing of dolphins in the Punjab Wildlife Protection Act in: **1974**
115. Since 2000, … has been working in collaboration with the Sindh Wildlife



Comp : ARHAM (+923351328979) | Dir: SAMUNA CH. | Co. : KAINAT CH.

116. Department on the conservation of the Indus River Dolphin. **WWF-Pakistan**
117. Male Saker Falcons usually have weight from: **730-990g**
118. Range Lifespan of Saker falcon: **10 Years**
119. … is the English name of Indian pangolin. **Scaly Ant Eater**
120. … are considered to be the most traded wild animal in the world. **Pangolins**
121. Para is the common name of: **Hog Deer**
122. Egyptian fruit bat is sometimes mistakenly called the: **Flying Fox**
123. The Egyptian fruit bat is a smaller type of bat, length of about: **6 Inches**
124. The snout area of Egyptian fruit bat looks very similar to that of a: **Canine**
125. Egyptian fruit bats can create very large colonies with often more than … members. **1000**
126. The anti-inflammatory drug … has been identified as the cause of mortality of Indian vulture. **Diclofenac**
127. SAVES stands for: **Saving Asia'S Vultures From Extinction**
128. … is the act of placing organisms into conditions that differ significantly from those experienced by these organisms in their natural habitats. **Release**
129. National Parks in Pakistan **29**
130. IUCN recognized … game reserves in Pakistan. **66**
131. The enclosures housing the endangered mammalian species have feeding & retiring cubicle/cell of: **Minimum Dimensions**
132. If any animal is found sick, injured or unduly stressed the matter shall be reported to the … for providing treatment expeditiously. **Veterinary Officer**
133. The zoo operators shall arrange for medical check-ups for the staff responsible for upkeep of animals at least once in every: **6 Months**
134. Any animal that dies in a zoo shall be subjected to a detailed: **Postmortem**
135. The findings recorded & maintained for a period of at least: **6 Years**
136. Every large & medium zoo shall make arrangements in the form of: **Recordings, Writings**
137. … are documents that are meant to guide decisions & to achieve certain goals. **Policies**
138. A safety policy should describe in detail what to do in case of animal escapes,

fire or other emergencies. **Acquisition & Disposition Policy**

139. A … creates a vision that is supported by policies, guidelines & priorities. **Master Plan**

140. The Director, Curator, Architects, veterinarians, Staff from keeper & Education departments are a part of: **Planning Team**

141. … gives details for the proposal & activities of development for next six years. **Management Plan**

142. Term used when there is no reasonable doubt that the last individual has died. **Extinct (Ex)**

143. When a taxon is known only to survive in Cultivation, Naturalized population well outside the past range: **Extinct In The Wild (EW)**

144. When a taxon is facing an extremely high risk of extinction in the wild in the immediate future: **Critically Endangered (Cr)**

145. Those which are abundant in their range but are gradually declining in total numbers. **Endangered (EN)**

146. When a taxon is not critically endangered or endangered but is facing a high risk of extinction in the wild in the medium-term future **Vulnerable**

147. When a taxon has not yet been evaluated against the criteria: **Not Evaluated (Ne)**

148. The IUCN Red List of threatened species lists … of internationally threatened animals occurring in Pakistan. **45**

149. Out of these 45 threatened species … are of mammals. **18**

150. Snow leopards have tails up to … in length which help the leopards to balance on steep and rocky slopes. **1m**

151. Unlike other large cats, snow leopards cannot: **Roar**

152. Body size of snow leopards can be up to … length weigh up to around: 1.3m, **70kg**

153. The gestation period of snow leopards is: **98-104 Days**

154. Snow leopard generally inhabits elevations b/w: **2000-4000m**

155. Estimated population of the snow leopard is b/w: **3,920-6,390**

156. There are as few as … approximately snow leopards in the wild: **6000**

157. Hyena's are large sized … carnivores. **Dog Like**
158. The ears of Hyenas lack any … on their outer margin. **Bursa Or Pouch-Like Fold**
159. In Pakistan, hyenas are distributed in Hilly tracks of: **Baluchistan & Sindh**
160. The … is a true desert dweller. **Sand Cat**
161. Protected areas are particularly adapted to the applications of … approaches. **Landscape**
162. Lal-Suhanra Park is located in: **Bahawalpur**
163. "GOL" means: **Valley**
164. … is used for skin care: **Salix Alba**
165. Worldwide there are now over … dams which alter the natural flow of water & impact on existing ecosystems**. 40,000**
166. Zoos reinforce the notion of … domination over non-human animals, which is never beneficial to animals. **Human**
167. Eagle, Hawks & their feathers have great cultural and spiritual value to the native of … as religious objective. **Americans**
168. Wild animals that can be hunted on an ordinary … with the details of everything is mentioned. **Permit**
169. Each zoo shall have wall at least … high from the ground level. **2m**
170. In Pakistan Lal Suhanra at Bahawalpur conserves: **Black Buck**
171. … on wildlife & necessity for the conservation should be screened even to the remotest villages & to all classes of students, so that they develop love and affection towards wildlife. **Cinematography**
172. Kingfisher have a … distribution. **Cosmopolitan**
173. … is not a scavenger. **Eagle**
174. About … % of domestic food crops now cultivated are descendants of wild plants. **90%**
175. The penalty of breaking wildlife act is imprisonment: which may extend to a period of: **1 Year**
176. The primary objective of habitat species management area is to maintain, conserve & restore specie and … **Habitat**
177. … is a common wet plant & is a staple diet of more than half of humanity. Rice



178. Which does not belong to Anatidae: **Houbara Bustard**
179. There are currently … specialist groups. **112**
180. … is the cause of species destruction. **Habitat Loss**
181. Introduction of … species is posing major threat to wetland. **Non-Native.**

SUBJECTIVES

Lectures 98 to 112

1. Maintenance of Zoo?

⇒ No zoo shall acquire any animal in violation of Wildlife Act.
⇒ No zoo shall exhibit any animal that is seriously sick or injured.
⇒ zoo shall use any animal, other than the elephant in plains and yak in hilly areas for riding purposes.
⇒ Each zoo shall be closed to visitors at least once a week.
⇒ Each zoo shall have a wall at least two meters high from the ground level.
⇒ No zoo shall have any residential complexes for the staff within the main campus for the zoo.
⇒ Such complexes, if any shall be separated from the main campus of the zoo by boundary wall with a minimum height of two meters from the ground level
⇒ No zoo shall keep any animal chained or tethered unless doing so is essential for its own well being
⇒ The zoo operators shall provide a clean and healthy environment in the zoo by planting trees, creating green belts and providing lawn and flowers bed etc.
⇒ The built-up area in any zoo shall not exceed twenty five percent of the total area of the zoo.
⇒ The built-up area includes administrative buildings, stores, hospitals, restaurants, visitors rest sheds, animal houses and pucca roads

2. Hygiene and Feeding Zoo Rules?

Maintenance of hygiene is important in every walk of life.

- Every zoo shall have ensured timely supply of wholesome and unadulterated food.
- The food must be in sufficient quantity to each animal according to the requirement of the individual animal, so that no animal remains undernourished.
- Every zoo shall have provided for a proper waste disposal system for treating both the solid and liquid wastes generated in the zoo.
- All leftover food items and animal excreta shall be removed from each enclosure regularly.
- Waste disposal should be done in a congenial manner to general cleanliness of the zoo.
- The zoo operators shall make available round the clock supply of potable water for drinking purposes in each enclosure.
- This would not be affecting the hygiene and cleanliness of zoo
- The large and medium zoo shall have an incinerator for disposal of the carcasses and other refuse material

3. What Is Animal Enclosure?

Essential features regarding animal enclosures are:

- All animal enclosures in zoo shall be so designed as to fully ensure the safety of:
    - Animals
    - Caretakers
    - Visitors,
- Stand of barriers and adequate warning signs shall be provided for keeping the visitors at a safe distance from animals.
- All animal enclosures in a zoo shall be so designed as the meet the full biological requirements of the animals housed therein.
- The enclosures shall be of such size as to ensure that the animals get space for their free movement and exercise.

- Within herds and groups animals are not unduly dominated by individuals

4. Mention the Number of Animal's Exhibits Large Zoo and Mini Zoo?

Number of the Animals exhibited

- Large Zoo (More than 750)
- Medium Zoo (500-750)
- Small Zoo (200-499)
- Mini Zoo (200)

5. Objective of Zoo?

Zoos are in a unique position to teach and inspire a diverse audience about wildlife and the role of zoos, the dangers facing animals in their natural habitats and to inspire visitors to get involved with conservation.

6. Administration Rule of Large Zoo?

General guidelines that zoo administration must follow for its maintenance

- No zoo shall acquire any animal in violation of Wildlife Act. No zoo shall exhibit any animal that is seriously sick or injured.
- No zoo shall use any animal, other than the elephant in plains and yak in hilly areas for riding purposes.
- Each zoo shall be closed to visitors at least once a week.
- Each zoo shall have a wall at least two meters high from the ground level.
- No zoo shall have any residential complexes for the staff within the main campus for the zoo. Such complexes, if any shall be separated from the main campus of the zoo by boundary wall with a minimum height of two meters from the ground level
- No zoo shall keep any animal chained or tethered unless doing so is essential for its own well-being. The zoo operators shall provide a clean and healthy environment in the zoo by planting trees, creating green belts and providing lawn and flowers bed etc.
- The built-up area in any zoo shall not exceed twenty five percent of the total area of the zoo.

- The built-up area includes administrative buildings, stores, hospitals, restaurants, visitors rest sheds, animal houses and pucca roads.

7. How Many Categories of Endangered Species?

Avoid products made from endangered species Overseas trips can be exciting and fun, and everyone wants a souvenir. But sometimes the souvenirs are made from species nearing extinction. Avoid supporting the market in illegal wildlife trade including Tortoise-shell, Ivory, Coral, Snakes, Turtles and lizards, live monkeys or apes. Be careful of products including Fur from tigers, polar bears, Crocodile skin, Medicinal products made from rhinos, tiger or Asiatic black bear, most like birds including Parrots, Macaws, Finches.

8. Record Maintain in Zoo...2 Marks

- No zoo shall acquire any animal in violation of Wildlife Act.
- No zoo shall exhibit any animal that is seriously sick or injured.
- No zoo shall use any animal, other than the elephant in plains and yak in hilly areas for riding purposes.
- Each zoo shall be closed to visitors at least once a week.
- Each zoo shall have a wall at least two meters high from the ground level.

9. What Is Zoo Boom?

In the 1950s, there was a 'Zoo Boom'. At that time entrepreneurs recognized the potential money to be made from exhibiting wild animals to satisfy public curiosity. Until the 20th century, there was little or no concern for the welfare of zoo animals. It was not until the late 1970s and early 1980s that the appearance of zoos began to change to reflect the public opinion. People were then educated about the true conditions endured by zoo animals

Lectures 113 to 127

10. Major Threats to Snow Leopard?

Major threats to snow leopards

- Poaching
- Conflict with communities
- Shrinking home:
- Changing climate
- Lack of effective law enforcement

I. Poaching

Snow leopards have long been killed for their Beautiful fur, Bones, Body parts, Traditional Asian Medicine.

II. Conflict with Communities

Retaliation for attacking herders' livestock. Decline in the leopard's natural prey is forcing them to rely more on livestock for food and increasing the risk of retaliatory killings. Due to hunting, competition from increasing livestock herds, and habitat loss.

III. Shrinking Home:

Snow leopards need vast areas to thrive. Expanding human and livestock populations are rapidly encroaching on their habitat. New roads and mines are also fragmenting their remaining range.

IV. Changing Climate

All the threats will be exacerbated by the impact of climate change on the fragile mountain environment. Hence, putting the future of snow leopards at even greater risk.

V. Lack of Effective Law Enforcement.

11. Threats to Wild Ass.

- Illegal trade seems to happen primarily on a national level.
- Poaching for meat, hides and fat, is believed to have apparent medicinal properties.
- Overgrazing by livestock reduces food availability, and herders also reduce the availability of water at springs. The cutting down of nutritious shrubs and bushes exacerbates the problem.
- A series of drought years could have devastating effects on this species

- Habitate fragmentation is a particular concern in Mongolia as result of the increasingly dense network of infrastructure.

12. Threats to Sand Dune Cat?

Threats

Habitat degradation and loss are considered to be the major threats to the Sand Cat. Many areas are also more heavily frequed by humans, their domestic animals and commensals (i.e., red fox, Vulpes.

- Arid ecosystems are being rapidly converted.
- Infrastructural development
- Human settlement and activity
- Degradation occurs through livestock grazing
- They are also killed in retaliation for killing chickens or Houbara and MacQueen's bustard

13. How Herbicides and Pesticides Can Affect Endangered Species?

Certain herbicides and pesticides are horrible pollutants that causing severe damage to the environment and animals dependent upon. To provide natural food to animal if spray is done to kill pest or un dew herbs it may cause problems to endangered species as they may die.

⇒ Find alternative ways to reduce negative effect on the wildlife in the area.
⇒ These are hazardous pollutants that affect wildlife at many levels.
⇒ Many herbicides and pesticides take a long time to degrade and build up in the soils or throughout the food chain.
⇒ Predators such as hawks, owls and coyotes can be harmed if they eat poisoned animals.
⇒ Some amphibians are particularly vulnerable to these chemical pollutants

14. Main Threat Panther?

Threats

- The primary threats to Leopards are anthropogenic.
- Habitat fragmentation
- Reduced prey base
- Conflict with livestock
- Game farming have reduced Leopard populations throughout most of their range
- Leopards are targeted for Trophy hunting
- Illegal wildlife trade for their Skins (used in traditional ceremonies) Bones and other parts (used for medicinal purposes in eastern cultures).
- In Indo-Malaya and China, a recent reports estimate very high levels of Leopard removal at four individuals per week for 10 years

Lectures 128 to 143

15. Biology and Behavior of Saker Falcon?

This bird is physically adapted to hunting close to the ground in open terrain, combining rapid acceleration with high maneuverability. Specializing on mid-sized diurnal terrestrial rodents (especially ground squirrels.

16. Feeding Habit of Indian Bustard?

The great Indian bustard is omnivorous feeding on seeds, crops, berries and cereals. They also feed on large insects, rodents, small birds and reptiles.

17. Macro Polo Sheep?

Scientific Name: Ovisammonpolii National Status: Critically Endangered IUCN Red list Threat Status: Near threatened Marco polo sheep is one of the nine Argali subspecies. The name ‘Argali' is the Mongolian word for wild sheep. It is the largest species of wild sheep. The general coloration varies between each animal. Light yellow to a reddish-brown to a dark grey brown. Argali from the Himalayas are usually relatively dark. Russian ranges argali are often relatively pale. The legs and belly are creamy white without any darker pattern on the frontal area.

18. Habitat of Hog Deer .3 Marks

Shrub dominated wetlands.Artificial terrestrial plantations like ChangaManga ,Riverine area, It prefers to live in grassy patches bordering forests.

19. Most Traded Wild Animal ?

Indian Pangolins are considered to be the most traded wild animal in the world. Leopard cat skins are commercially traded internationally for the fur trade.

20. Not on Indus Dolphin 3

Description

It has a long beak which thickens towards the tip, large teeth. Body is stocky with rounded belly. Flippers are large and paddle shaped. Forehead is steep. Poorly seeing eye

Eye hasn't developed a lens; the dolphin still uses its eyes to differentiate between light and dark. Tail flukes are broad in relation to body size. Grey brown in color, sometimes with a pinkish belly. Males smaller than females. Length 1.5 and 2.5m.

Habit & Habitat

These dolphins favor silt laden, turbid waters of the Indus river system. At temperature 8C° and 33C°. Deepest river channel. Deep, low-velocity water. Less common in secondary channels and small braids

Distribution Native

- India
- Bangladesh
- Nepal
- Pakistan

Threats

- Human development
- The construction of dams and dikes segments the populations reducing the mature individuals to mate
- It also reduces their habitat because they cannot move freely through these structures.



21. 3 Zoo Rules?

Zoo is a facility in which animals are

- Housed within enclosures
- Displayed to the public
- They may also breed
- "Today, the zoo is on the list of inevitable field trips and vacation destinations."

22.Why Animal Trapped

United Nations Development Programmed (UNDP) to rescue dolphins trapped in irrigation canals and return them to the Indus mainstem has had some success in reducing mortality.

23.Threats Indian Pangolin.

English Name: Scaly Ant eater
Scientific Name: Manis crassicaudata
National Status: Vunerable
IUCN Red list Threat Status: Endangered

Pangolins are considered to be the most traded wild animal in the world.

Description

The Indian Pangolin has Relatively tiny head, Hump-backed body. Thick tapering tail almost equal in length to its body. They are completely covered with scales. Made of keratin which is getting harden as the creatures get older. The scales bear fine longitudinal striation on their surface. There is hardly any external ear. The muzzle tapers to a narrow, down-curving snout. The hind legs are stout and rather columnar with five blunt pinkish white toe nails.

Habitat and Ecology

It occurs in various types of tropical forests as well as open land, grasslands. Degraded habitat, including in close proximity to villages. The species is thought to adapt well to modified habitats. Provided its ant and termite prey remains abundant.

The animal is Solitary, Nocturnal, Burrow-dwelling. During mating season, adult males and females share the same burrow. Females usually give birth to one young. Gestation period of 165 days. Longevity in the wild is unknown. In captivity it has been recorded up to 13 years 2 months. This species is arboreal in some habitats, and is a good climber, using its prehensile tail and claws to climb trees Distribution

Native:

India, Nepal, Pakistan Mangla, Potohar and Salt Range across Indus south to near Karachi and northward in the mountainous areas uptoKallat

- Sri Lanka
- Possibly extinct:
- Bangladesh

Threats

Primarily threatened by Hunting, Poaching (for meat and scales), Illegal international trade. Up to 100,000 pangolins are estimated to be hunted and sold every year its meat is consumed as a source of protein locally.

The meat is considered a delicacy in China, Vietnam and other parts of south-east Asia. Scales are used in whole or powdered form in the preparation of traditional medicines. An increase in the Agrarian economy Improved irrigation. The use of pesticides comprises additional threats to this species. Occasionally killed out of fear for having an odd shape

24.Indian Spotted Eagle Description?

Description

A medium-sized eagle about 60 cm in length and has a wingspan of 150 cm with a short tail. Adults are essentially brown. The legs appear longer and thinner due to the tarsi being less thickly feathered. This species has a lighter coloration overall compared to its relatives with a darker iris that makes the eyes appear darker than the plumage. The young birds are giving a spotted appearance glossy brown. Tips of the head and neck feathers being creamy. The upper tail coverts are light brown with white giving a barred appearance. The median coverts have large creamy

spots.After about eighteen months the bird molts and becomes a darker shade and has less spots.

Lectures 144 to 159

25.Indian Pangolin Native Countries?

India, Nepal, Pakistan

26.Extent Animals Names 2 Present in Pakistan?

The primary threat to pangolins in Asia is poaching and illegal hunting for illicit international trade, predominantly for pangolin meat, which is considered a delicacy in China and Vietnam, and for pangolin scales, which are used as an ingredient in traditional Asian medicine.

27.Threat to Indian Vultures?

Threats

These vultures were being found dead and dying in Pakistan and throughout India. The anti-inflammatory drug diclofenac has been identified as the cause of mortality. Vultures are exposed to diclofenac and other NSAIDs through scavenging on the carcasses of largely cattle and buffalo that have been treated with the drugs prior to death and left for scavengers to consume. Hence likely contributory factors are

- Poisoning
- Pesticide use
- Conversion and disturbance of forested habitats within its range.

28.Write the Short Notes On Indian Vulture?

The Indian vulture (Gyps indicus) is an Old-World vulture native to India, Pakistan and Nepal. It has been listed as Critically Endangered on the IUCN Red List since 2002, as the population severely declined. ... It breeds mainly on hilly crags in central and peninsular India.

Lectures 160 to 173

29. Explain Urial And Its Sub Types?

Urial is a wild sheep with long legs and relatively small horns, native to central Asia. Urial is represented by three sub species in Pakistan. Ovisammon (Marcopolo sheep) Ovisorientalis (Urial)

30.Feeding Habitat of Barking Deer?

Diet includes grasses, low growing leaves, tender shoots and fruits.

31. Short Note on Brown Bear

Ursus arctos Grizzly Bear

Description:

- Large animals
- Dark in colour
- Long guard hair over the shoulder
- Characterized by adistinctive humps on shoulders
- Long claws on the front paws
- Sexual maturity: 4.5 to 7 years of age.
- Litter size ranges from one to four but two are common.
- Cubs remain with their mothers for at least two and half years.
- Female breed every three years.

Feeding Habit:

- Chiefly vegetarian
- Feed mostly on fruits, honey and ants.
- Occasionally kill sheep and goat.

Threats

- Brown Bears are attracted to areas with available human-related foods
- Being large and somewhat aggressive, these bears may threaten life and

property (often agricultural products) and may be killed as a consequence.

Conservation Actins

Conservation actions vary greatly among nations and regions within nations. Large populations of this species are legally hunted, and thus managed as a game animal.

32. Capra Description

- Body length 52 inches, 4.4 feet
- Have strong limbs terminating in broad hooves.
- Female and young male
- Yellow brown to reddish grey
- Male are beautiful with long horn
- White hair in old age
- Male have short beard
- Female black beards
- Gestation period: 150-155 days, 5 months
- Twins are common
- Sexual maturity: About 3 years
- Gregarious from large herds.
- Older male generally keeps together
- Wild goat wonderful has sense of balance and can make standing leap upwards.
- Food: The leaves and bushes small shrubs and herbs.
- Within herds and groups animals are not unduly dominated by individuals

33. Leopard Cat Description.

Scientific Name: Prionailurusbengalensis
IUCN Red list Threat Status: Least concern

Description

The leopard cat looks like a miniature, long-legged and more slender version

of a leopard. Body markings are like leopard; head with stripes.
Its fur varies highly in color and markings according to the region. Generally spotted body fur coat

- In the tropics Fur is yellowish-brown with white underparts
- In the northern part of the species' range Greyish brown thick fur and are also larger and heavier than the ones in the south.

The size and shape of the black markings covering the body and limbs are also highly variable. The spots sometimes form lines along the neck and back. The irises are deep, golden brown to grey. Males are larger than females

- Body length 54-60 cm
- Weight 3-4 kg.
- Longevity: up to 13 years
- Litter size:2-3 cubs

Habitat and Ecology

Leopard Cats can occur in a wide variety of habitats. Tropical rainforest, Temperate coniferous forest, Shrub forest and successional grasslands. The northern boundary of its range is limited by snow cover. Leopard Cat does not occur in deep snow. It is not found in the cold steppe grasslands, and generally does not occur in arid, treeless areas.The Leopard Cat is predominantly nocturnal and crepuscular. Some diurnal activity has been recorded.

Distribution

The leopard cat occurs throughout most of India, West into Pakistan and Afghanistan. Its range extends into the Himalayan foothills across most of China. North to the Korean peninsula and into the Russian Far East.It is found in most of Southeast Asia: Thailand, Vietnam and China. On the islands of Sumatra, Java, Borneo and Taiwan. On small islands off South Korea. The leopard cat is the only wild cat species which is native to Japan and the Philippines.

Threats

Leopard cat skins are commercially traded internationally for the fur trade,

primarily coats.Skins are also used as decorations in some parts of its range. They are occasionally kept as pets. Sometimes taken from the wild. Sometimes from captive breeding. They have been interbred with domestic cats, particularly in the West, to make the popular Bengal breed. Their bones are used in some traditional Asian medicines

Conservation

Leopard Cat populations in Bangladesh, India and Thailand are listed on CITES Appendix I. All other populations are listed on Appendix II.

34.Note On Barking Deer?

Muntjacs

- Muntiacusmuntja
- Cervusmoschatus
- Cervusmuntjak

IUCN Red list Threat Status: Least concern It gives calls similar to barking, usually upon sensing a predator. Hence the common name for all muntjacs of "barking deer

Description:

It is a little dainty creature and weighs about 22-25 kg. Coloration varies from deep brown to yellowish with creamy or whitish markings. Only males have antlers. The upper canines of the male are elongated into tusks, which curve strongly outward from the lips. Females also develop small tusks in the upper jaw. Capable of inflicting sever injuries with these tusks, when defending themselves or their fawns. Two bony ridges are present in the front part of the skull.

Feeding habit:

Diet includes grasses, low growing leaves, tender shoots and fruits.

Distribution

Native: Brunei Darussalam; Indonesia; Malaysia; Thailand

Regionally extinct: Singapore

Pakistan: Confined to Himalayan foothills. Commonly found in Margala hills, Swat and Azad Kashmir.

Threats

Generally, Muntjac is an important source of bushmeat. Selected body parts and the foetus are also used in traditional medicine. Individuals are usually killed during, or shortly after, their capture. Foothold snares are commonly used as well as spears and firearms. Dogs are often employed.

35. Note On Capra Hircus?

Capra Hircus

Females are beardless but carry backward curving horns. Measure up to 15 cm and bear fine annulations towards their base. The younger males and females show a characteristic shape pattern with a dark stripe running from the corner of eye to the muzzle. Mature males are spectacularly beautiful and may have horns over one meter in length. In adults the hair coat is of reddish buff in color. The belly and the outside of lower limbs, beard and fore part of face varies from black to chestnut brown in color. The shoulder height. They are capable of surviving in very arid desert conditions. They inhabit mountain crests up to 3300 m.

- In males is 85 to 95 cm. In females 55 to 60 cm.
- The weight of adult Males up to 120 kg, Females 50 to 55 kg.

36.Capra Description?

Description:

Body length 52 inches, 4.4 feet have strong limbs terminating in broad hooves. Female and young male yellow brown to reddish grey. Male are beautiful with long horn, white hair in old age. Male have short beard, Female black beards.

Lectures 174 to 187

37.Marten Are Found In Pakistan?

Stone martens are found throughout mountain ranges of Baluchistan and Northern areas.

38.Types of Marten

- Two species common to Pakistan are
- Stone marten (Martes foina)
- Yellow throated marten (Martes flavigula)
- The stone marten has a dense fur coat of rich liver chestnut hue.

39.Scientific Name of Domestic Goat?

Capra falconeri

40.Feeding Habits of Common Peafowl?

Feeding habit Omnivorous:

- Grain and green crops to insects
- Small reptiles
- Mammals
- Berries
- Termite
- Grasshopper
- Ants and beetles.
- Roots in high fairly open trees in large number.

Peafowl have an omnivorous diet, that is, they will eat both other animals and plants. They feed from the soil - they don't fly - and that is where they take most of their nutrients. ... Among their favorite foods we can point out fruits, seeds, berries, plants, vegetables, insects, larvae and small reptiles.

41.Scientific Name of Wild Goat and Domestic Goat?

BLACK BUCK ……. Antilope cervicapra
WILD GOAT…. Capra aegagrus.

42.Scientific Name of Black Buck?

Antelope Cervicapra.

43.How Many Species of Martens?

Two species common to Pakistan are

- **Stone marten (Martesfoina)**

The stone marten has a dense fur coat of rich liver chestnut hue

Distribution

Stone martens are found throughout mountain ranges of Baluchistan and Northern areas.

- **Yellow throated marten (Martesflavigula)**

The yellow throated marten is considerably larger in size than the stone marten. The whole of its chest, throat and lower jaw are yellowish in color. General body color is silvery, tail is longer; rhinarium is naked and black.

Distribution

Yellow throated marten is found in Chitral, Azad Kashmir and Margala hills

Lectures 188 to 201

44.Characteristics of Male Tragopan.5

Facts:

Tragos = goat

Pan = Greek god

Tragopans are commonly called "horned pheasants" Having two brightly colored, fleshy horns on their heads. They can erect during courtship displays. There are five recognized species of tragopans.

- Western tragopan (Tragopan melanocephalus)
- Satyr tragopan (Tragopan satyra)
- Timmins's (tragopan Tragopantem minckii)
- Blyth's tragopan (Tragopan blythii)
- Cabot's tragopan (Tragopan caboti)

Tragopanmelanocephalus

Syn: Western Tragopan, Black-headed Tragopan, Western Horned-pheasant

Description:

Medium sized pheasant. Sexes are highly dimorph. Male grey upper parts and reddish facial skin Numerous white spots. Each spot bordered with black and deep crimson patches on the sides and back of the neck. The throat is bear with blue skin while the bare facial skin is red. They have a small black occipital crest.

- Weighs 1.8–2.2 kg (4.0–4.9 lb.)
- Length from 55–60 cm (22–24 in)

Females have pale brownish-grey upper parts finely vermiculated. Spotted with black, and most of the feathers have black patches and central white streaks.

- Weigh 1.25–1.4 kg (2.8–3.1 lb).
- Length 48–50 cm (19–20 in).
- Monogamous

Feeding Habit: The diet primarily consists of leaves of trees and shrubs. Also includes roots, flowers, acorns, seeds, berries, grubs and insects. Tragopans are shy and wary birds quick to hide amongst the dense vegetation they inhabit if disturbed.

Distribution

Native: India, Pakistan

Found in western Himalayas, between 8000-12000 from hazara eastward to Garhwali- Ladakh

IUCN Status: Vulnerable

Distribution in Pakistan

- Largely restricted to area between Jhelum and Kunhar rivers of Hazara district
- Forest of Kaghan valley, in Machiara National park (A.K)
- Pales valley in Cholistan district

Threats

Hunting and trapping for

- Its meat (especially in winter)
- Its decorative plumage

45. What Are Conservation Strategies for King Fisher?

Conservation Actions Underway

- Bern Convention Appendix II.
- EU Birds Directive Annex I.
- There are no known current conservation measures for this species within its European range.

Conservation Actions proposed

- Preservation of rivers and streams must be mediated through industry, agriculture and water Authorities
- Research to investigate the importance of individual threats and mitigation measures against all the factors.

46.Habitat of Chukor?

Habitat

Lives on barren rocky slopes with sparsely stunted grass and bushes.In winters, it comes down to an elevation of 1200-1500 m. Re ascends in summer to 2500 m.

FROM MID TERM

47.National Park?

Large natural or near natural areas set aside to protect large-scale ecological processes, which also provide a foundation for environmentally scientific, educational, recreational and visitor opportunities. Primary objective to protect natural biodiversity along with its underlying ecological structure.

Other Objectives:

- ⇒ To manage the area in order to perpetuate, in as natural a state as possible. To maintain viable and ecologically functional populations.
- ⇒ To contribute to local economies through tourism.
- ⇒ To manage visitor, use for inspirational, educational recreational purposes. Distinguishing features
- ⇒ The area should contain representative examples of major natural regions, and biological and environmental features or scenery.
- ⇒ It should be of sufficient size to maintain ecological processes.
- ⇒ The composition, structure and function of biodiversity should be to a great degree in a “natural” state.

48.IUCN Definition of National Park

Large natural or near natural areas set aside to protect large-scale ecological processes, which also provide a foundation for environmentally scientific, educational, recreational and visitor opportunities.

49.Game Reserves In Pakistan

There are 96 Game reserves in Pakistan IUCN recognized 66 Game/Hunting reserves in Pakistan.

50.What Is Wildlife

"Wildlife" refers to neither cultivated or domesticated nor tamed organisms. Or Organisms that grow or live wild in an area without being intervened by humans. Wild life is natural renewable resource with a great importance throughout world. It comprises innumerous varieties of living organism in their natural habitats.

- Wild Plants
- Wild Animals
- Fungi Microorganisms

Wildlife can be found in all ecosystems.

- Deserts
- Forests
- Rain forests
- Plains
- Grasslands

51. 4 National Park

- Kirthar National Park
- Hingol national park
- Lal Suhanra National Park
- Hazarganji Chiltan National Park
- Ayubia National Park
- Khunjerab National Park

52.Two Names Of Dams

- Mangla dam
- Terbela dam

53.Game Reserve

Definition:

"A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis" A game reserve (wildlife preserve) is a large area of land where wild animals live safely or are hunted in a controlled way for sport.

⇒ In the game reserves the major focus is specifically the animals.
⇒ If hunting is prohibited, a game reserve may be considered a" Nature Reserve"
⇒ Wherein all aspects of naturally-occurring life in the area are considered.
⇒ Most of the areas in game reserves have created to provide habitat protection for animal species commonly referred to as game (huntable species for sport or meat)

54.How Many Wildlife Sanctuaries in Pakistan?

In Pakistan presently there are 99 Wildlife Sanctuaries.

55.Name the Largest National Park of Pakistan

Hingol National Park is the largest in Pakistan

56.Hingol National Park Location

It is located along the Makran coast in southwestern, Balochistan-Pakistan.

57.Why Hunting of Markhor Performed

Markhor hunting for trophy

58.Location of Cholistan Wildlife Sanctuary

♥ Cholistan desert
♥ In the South Eastern portion of Punjab

59.Location of Margalla Hills National Park

North of the Islamabad City, Capital of Pakistan.

60. Note on Ramsars Of Sindh

⇒ Keenjhar (Kalri) Lake
⇒ Haleji Lake
⇒ Drigh Lake
⇒ Indus Dolphin reserve
⇒ Jubho lagoon
⇒ Nurri Lagoon

61. Aesthetic Important Of Wildlife?

It is the pleasure to enjoy the beauty of wildlife

- People enjoy seeing plants and animals in their natural habitats for recreation.
- They are a source of joy, wonder, and pleasure for many people.
- Wildlife tourism (ecotourism).
- Bird watching, is now a growing part of wildlife tourism.
- Animals photographed for recreational purposes also have aesthetic value.
- Exotic game ranches
- Viewing non-native species in a "near-native" habitat.
- Visiting Safari parks/zoo as recreational activity
- Visiting hilly areas for recreational purpose

62. Wetland Types

Types of Wetlands: Five major wetland types are generally recognized.

- Marine
- Estuarine
- Lacustrine
- Riverine
- Palustrine

Marine: Coastal wetlands including coastal lagoons, rocky shores, and coral reefs.
Estuarine: Including deltas, tidal marshes, and mangrove swamps.

Lacustrine: Wetlands associated with lakes.
Riverine: Wetlands along rivers and streams.
Palustrine: Wetlands that lacks flowing water, "marshy";marshes, swamps and bogs.

63.Species Conserved By SGP?

Small Grants Programme (SGP)

"Small support for long term impact".

SGP of WWF-Pakistan has been supporting grass root level organizations and researchers to strengthen efforts to conserve threatened and endangered species and ecosystem. Under the programme more than 450 projects have been successfully completed.

SGP has supported innovative conservation initiatives for priority species and ecosystem.

- Marine turtles
- Snow leopard
- Indus River dolphin
- Sharks
- Mangroves forests
- Indus river delta
- Coniferous forests.

Collaborative research programs should be initiated with neighboring countries with common ecosystem.

64. Note on Rabies?

Rabies (Hydrophobia): Bite Wound Disease
Host: Any mammal, usually carnivores; highest incidence is in skunks, foxes, bats, and raccoons.
Transmission: Bite wounds, infected saliva in cuts or skin abrasions, aerosol in bat caves.



65. Name National Park?

- Ayub National Park
- Chinji National Park
- Hingol National Park

66. Three Rules for Zoo?

Zoo Rules

Zoo is a facility in which animals are

- Housed within enclosures
- Displayed to the public
- They may also breed
- "Today, the zoo is on the list of inevitable field trips and vacation destinations."
- Every zoo shall have one full-time officer in charge of the zoo.
- The said officer shall be delegated adequate administrative and financial powers as may be necessary for proper upkeep and care of zoo.
- Stand of barriers and adequate warning signs shall be provided for keeping the visitors at a safe distance from animals.
- All leftover food items and animal excreta shall be removed from each enclosure regularly

67. Primary Objective of National Parks?

Primary objective

To protect natural biodiversity along with its underlying ecological structure.

Other Objectives

⇒ To manage the area in order to perpetuate, in as natural a state as possible.
⇒ To maintain viable and ecologically functional populations.
⇒ To contribute to local economies through tourism.

⇒ To manage visitor use for inspirational, educational recreational purposes

68. Location of Safari Park?

Safari park in Lahore Pakistan.

69. Experimental Wild Animals?

Wild animals as Experimental Animals - Monkeys, Rabbits, Guinea-pigs, Rats etc.

70. Define Wildlife?

"Wildlife" refers to neither cultivated or domesticated nor tamed organisms. Or Organisms that grow or live wild in an area without being intervened by humans.

Wild life is natural renewable resource with a great importance throughout world.

71. Name Largest National Park of Pakistan?

Hingol National Park was established in 1988. It is now named as Dharune-Hingol National Park, formed by the emergence of two Parks.

72. Onus Of Proof?

Onus of Proof

- When in any proceedings taken under this Act.
- Wild animal, trophy or meat shall be presumed to be the property of until the contrary is proved.

73. Mintaka Means?

Mintaka means: "Thousand goats.

74. How Birds And Insects Help To Maintain Ecosystem?

Insects and birds add to the economic gains through pollination of plants.

Maintaining the 'balance of Nature'. e.g., Carnivore-herbivore balance.

75. Define Territory.

It is basically the home of the wild animals, it's the particular area in which animals itself defined by other animals or may within the same species to fight to protect that particular area that is called Territory.

76. What Is Location of Astore Sanctuary?

Location:

- Astore District
- Between Nanga Parbat
- Gilgit–Baltistan, Pakistan

77. How Can We Educate People About Wildlife Protection?

Most people of any country are ignorant about the utility of wildlife. Hence, for the conservation of wildlife there is a great need of educating people.

To achieve this goal following methods can be adapted.

- Celebration of wild life week every year.
- Publicity through media and film shows.
- Holding conducted tours, essay competitions, lectures, seminars, etc.
- Setting up nature clubs in educational institutions.
- Publication of wildlife books and journals.
- Establishment of natural history museums etc.

78. Aims Of PAWS?

Pakistan Animal Welfare Society (PAWS)

Aims to create an equitable relationship between humans and animals in Pakistan.

79. How Pollution Effect Wetlands

Wetland degradation is a big and growing problem occurring due to pollution inputs.

The primary pollutants causing wetland degradation are

- Sediments
- Fertilizers
- Human sewage
- Animal waste
- Road salts
- Pesticides
- Heavy metals

80. 2 Wild Sanctuaries of KPK?

- Manglot
- Argam Basti

81. Define Wetlands.

Wetlands are areas where water is the primary factor controlling the environment and the associated plant and animal life.

They occur where the water table is at or near the surface of the land, or where the land is covered by shallow water.

82. Trading of Animals?

Trading of wildlife and its products'. Bush meat, Ivory, hide, trophy, fur etc.

83. Invasive Species?

An invasive or alien species is established outside of its natural past or present distribution, whose introduction and/or spread threaten biological diversity.

An invasive species is a species that is not native to a specific location (an

introduced species), and that has a tendency to spread to a degree believed to cause damage to the environment, human economy or human health.



"برادوست کوئلے کی مانند ہے،
اگر جلتا ہو گا تو آپ کا ہاتھ جلادے گا
اگر بجھا ہو گا تو آپ کے ہاتھ کالے کر دے گا"

~SAMUNA~